Taylor McDonald

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Data Science student at NJIT building ML-powered full-stack applications in predictive modeling and real-time data systems. Seeking a Summer 2026 internship in data science or ML engineering.

SKILLS

Programming Languages: Python, SQL
ML/DS: scikit-learn, XGBoost, pandas, numpy, PyTorch (basics)
Tools: Git, FastAPI, Jupyter, VS Code, Hugging Face Spaces
Areas: Machine Learning, Statistical Learning/Modeling, Regression Analysis, Data Wrangling, Model Deployment
Spoken Languages: English(fluent), Korean (Intermediate)

PROJECTS

PawsIQ — Pet Services Intelligence Platform (In Progress) | Github **April 2026 – Present**

- Developing demand forecasting model using XGBoost on synthetic booking data to predict peak walk request times by hour and zip code
- Engineering dynamic pricing pipeline using Ridge Regression to adjust service rates based on real-time demand signals, targeting measurable revenue lift over flat-rate baseline
- Designing PostgreSQL schema and FastAPI backend to serve ML predictions to a React dashboard; building visualizations for churn risk scores and walker sentiment trends across 20+ simulated clients
- Stack: Python, XGBoost, scikit-learn, FastAPI, PostgreSQL, React

House Prices Prediction | Github **March 2026**

- Engineered 5 features (TotalSF, TotalBathrooms, HouseAge, OverallScore, binary flags) from 79 raw variables using Random Forest importance, reducing noise and improving model signal
- Trained and compared 3 models (Linear Regression, KNN, Random Forest) on 1,460 samples, achieving R-squared=0.87 and RMSE=0.14 with Random Forest - a 22% RMSE improvement over the Linear Regression baseline
- Cleaned and imputed missing values across 79 features, producing a fully reproducible pipeline from raw Kaggle CSV to final predictions
- Stack: Python, pandas, numpy, scikit-learn, matplotlib, seaborn

FocusFit (In Progress) | Github **Dec 2025 - Present**

- Developing ML model to predict optimal study session lengths from user focus pattern data
- Engineered features from session logs using pandas; evaluated classifier performance via accuracy and F1 score
- Visualized productivity trends with matplotlib to surface actionable insights
- Stack: Python, pandas, scikit-learn, matplotlib

WORK EXPERIENCE

All The Paws | Founder | East Orange, NJ **Sep 2025 - Present**

- Founded and registered a pet services business, growing to 15+ recurring clients within months by building trust through consistent, high-quality service delivery
- Designed and implemented end-to-end booking and scheduling systems - covering invoicing, daily care routines, and emergency protocols - reducing operational friction across all client accounts
- Built and maintain the business website, managing branding, service copy, and client-facing UX to support client acquisition

EDUCATION

Data Science B.S. | New Jersey Institute of Technology, Newark, NJ **Expected May 2027**
GPA: 3.1 | Sep 2024 - Present
Coursework: Intro to ML, Regression Analysis, Statistical Methods I and II, Statistical Methods in Data Science, Multivariate Distribution, Linear Algebra

Study Abroad | Hanyang University ERICA, Ansan, South Korea **Feb 2025 - Jun 2025**
Coursework: Advanced Calculus, Intensive Korean

HONORS AND ACTIVITIES

JPMorgan Chase and Co. Career.edYOU Academy Graduate | 2023
Headstarter AI Software Engineering Fellow | 2024
Community Service Certificate | Mission Trip, South Korea | 2024
Onyx Dance Crew | NJIT | 2025

CERTIFICATIONS

Python Certification (In Progress) | freeCodeCamp **January 2026**
Supervised Machine Learning: Regression and Classification | Coursera **February 2025**
Foundations: Data, Data, Everywhere | Coursera **May 2024**